



# Multiplying and Dividing Radicals: Adding, Subtracting, Multiplying, Dividing

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## Learning Objectives

- Simplify radical expressions and identify like radicals
- Add and subtract radical expressions by combining like terms
- Multiply radical expressions using the product rule
- Divide radicals and rationalize denominators

*Simplify each radical expression completely and rationalize all denominators where needed.*

### 1. Simplify the radical expression.

$$\sqrt{72}$$

Answer: \_\_\_\_\_

### 2. Identify and combine the like radicals.

$$5\sqrt{3} + 2\sqrt{3} - \sqrt{3}$$

Answer: \_\_\_\_\_

### 3. Add the radical expressions by first simplifying.

$$\sqrt{50} + \sqrt{18}$$

Answer: \_\_\_\_\_

### 4. Subtract the radical expressions by simplifying each first.

$$3\sqrt{75} - 2\sqrt{12}$$

Answer: \_\_\_\_\_

### 5. Multiply the radicals and simplify.

$$\sqrt{6} \cdot \sqrt{15}$$

Answer: \_\_\_\_\_

### 6. Multiply the binomial radical expression and simplify.

$$(2\sqrt{3} + 1)(\sqrt{3} - 4)$$

Answer: \_\_\_\_\_



**7. Divide the radicals and simplify.**

$$\frac{\sqrt{48}}{\sqrt{3}}$$

Answer: \_\_\_\_\_

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**8. Rationalize the denominator and simplify.**

$$\frac{5}{\sqrt{7}}$$

Answer: \_\_\_\_\_

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**9. Rationalize the denominator and simplify.**

$$\frac{6}{\sqrt{12}}$$

Answer: \_\_\_\_\_

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**10. Rationalize the denominator using the conjugate.**

$$\frac{4}{\sqrt{5} + 1}$$

Answer: \_\_\_\_\_





This worksheet covers all topics from the video: review of radical expressions, combining like terms with radicals, adding and subtracting radicals, multiplying radicals (including binomial products), dividing radicals using the quotient rule, and rationalizing fractions with single-term and binomial denominators.

## Solutions

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1. Simplify the radical expression.

$$\sqrt{72}$$

- Factor 72 as 36 times 2, where 36 is a perfect square.
- Take the square root of 36 to get 6.
- Write the simplified form as 6 times the square root of 2.

**Answer:**  $6\sqrt{2}$

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2. Identify and combine the like radicals.

$$5\sqrt{3} + 2\sqrt{3} - \sqrt{3}$$

- All three terms have the same radical part, the square root of 3, so they are like radicals.
- Combine the coefficients: 5 plus 2 minus 1 equals 6.
- Write the result as 6 times the square root of 3.

**Answer:**  $6\sqrt{3}$

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3. Add the radical expressions by first simplifying.

$$\sqrt{50} + \sqrt{18}$$

- Simplify the square root of 50 as 5 times the square root of 2.
- Simplify the square root of 18 as 3 times the square root of 2.
- Add the like radicals: 5 plus 3 equals 8, giving 8 times the square root of 2.

**Answer:**  $8\sqrt{2}$

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4. Subtract the radical expressions by simplifying each first.

$$3\sqrt{75} - 2\sqrt{12}$$

- Simplify the square root of 75 as 5 times the square root of 3, so 3 times that gives 15 times the square root of 3.
- Simplify the square root of 12 as 2 times the square root of 3, so 2 times that gives 4 times the square root of 3.
- Subtract the like radicals: 15 minus 4 equals 11, giving 11 times the square root of 3.

**Answer:**  $11\sqrt{3}$

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5. Multiply the radicals and simplify.

$$\sqrt{6} \cdot \sqrt{15}$$

- Use the product rule to combine into the square root of 6 times 15, which equals the square root of 90.
- Factor 90 as 9 times 10, where 9 is a perfect square.
- Take the square root of 9 to get 3, leaving 3 times the square root of 10.

**Answer:**  $3\sqrt{10}$

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6. Multiply the binomial radical expression and simplify.

$$(2\sqrt{3} + 1)(\sqrt{3} - 4)$$

→ Use FOIL: first terms give 2 times 3 equals 6, outer gives negative 8 times the square root of 3.

→ Inner gives the square root of 3, and last gives negative 4.

→ Combine constants 6 minus 4 to get 2, and combine radicals negative 8 plus 1 to get negative 7 times the square root of 3.

**Answer:**  $2 - 7\sqrt{3}$

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7. Divide the radicals and simplify.

$$\frac{\sqrt{48}}{\sqrt{3}}$$

→ Use the quotient rule to write this as the square root of 48 divided by 3.

→ Simplify 48 divided by 3 to get 16.

→ Take the square root of 16 to get 4.

**Answer:**  $4$

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8. Rationalize the denominator and simplify.

$$\frac{5}{\sqrt{7}}$$

→ Multiply the numerator and denominator by the square root of 7.

→ The numerator becomes 5 times the square root of 7.

→ The denominator becomes the square root of 49, which equals 7.

**Answer:**  $\frac{5\sqrt{7}}{7}$

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9. Rationalize the denominator and simplify.

$$\frac{6}{\sqrt{12}}$$

→ Simplify the square root of 12 as 2 times the square root of 3, giving 6 over 2 times the square root of 3, which simplifies to 3 over the square root of 3.

→ Multiply numerator and denominator by the square root of 3 to get 3 times the square root of 3 over 3.

→ Cancel the 3 in the numerator and denominator to get the square root of 3.

**Answer:**  $\sqrt{3}$

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10. Rationalize the denominator using the conjugate.

$$\frac{4}{\sqrt{5} + 1}$$

→ Multiply the numerator and denominator by the conjugate, the square root of 5 minus 1.

→ The numerator becomes 4 times the quantity square root of 5 minus 1.

→ The denominator becomes 5 minus 1, which equals 4, and then the 4 in the numerator cancels with the 4 in the denominator.

**Answer:**  $\sqrt{5} - 1$

